

CSE4261-DNN Assignment-2

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1 Effect of Activation Functions in CNN Models

[Github code link](#)

The choice of activation function plays a critical role in the performance and convergence behavior of convolutional neural networks (CNNs). This section presents a comparative overview of ten widely used pre-trained CNN architectures, highlighting the activation functions integrated within each model and their respective computational effects. The table describes effects of using activation functions.

Table 1: Effect of ReLU and Sigmoid activation in pretrained CNNs

Model	ReLU	Sigmoid / Similar
Xception	Fast, efficient in depthwise layers.	Sigmoid not ideal.
VGG16	Common after conv layers, speeds training.	Sigmoid may slow learning, rarely used.
VGG19	Supports deep training with stable gradients.	Sigmoid activation can saturate early layers.
ResNet50	Works well with residuals and BN.	Sigmoid causes gradient issues in deep nets.
ResNet50V2	Pre-activation ReLU improves flow.	Sigmoid rarely used due to slow convergence.
ResNet101	Maintains gradient in deep layers.	Sigmoid tends to vanish gradients in depth.
ResNet101V2	ReLU enhances pre-activation learning.	Sigmoid unsuitable for deep residual blocks.
ResNet152	ReLU helps deeper propagation.	Sigmoid not effective in deep models.
ResNet152V2	Pre-activation ReLU aids training.	Sigmoid less used. may harm gradient flow.
MobileNet	ReLU6 helps low-power inference.	Sigmoid not efficient for mobile devices.

2 CNNs Using Different Types of Kernels

Kernels are the core components of convolutional layers in CNNs, responsible for feature extraction from images. Different types of kernels have been developed to improve performance, efficiency, and adaptability in various tasks. From table2, we get examples of CNN models that utilize regular, deformable, dilated, depthwise separable, modified depthwise separable, and pointwise kernels, each offering unique advantages in terms of accuracy and computational cost.

Table 2: Different types of kernel used by CNNs

Kernel Type	CNN Models
Regular Kernel	VGG16, ResNet50, AlexNet.
Deformable Kernel	Deformable ConvNets, Deformable ResNet, Deformable Faster R-CNN.
Dilated (Atrous) Kernel	DeepLabv3, Dilated Residual Networks (DRN), WaveNet.
Depthwise Separable Kernel	MobileNetV1, Xception, EfficientNet.
Modified Depthwise-Separable Kernel	MobileNetV2, ShuffleNet, EfficientNet.
Pointwise Kernel	ResNet, MobileNet, Inception-v3.

3 Feature Map Analysis of VGG16

- **Input Layer:** Accepts images of shape $32 \times 32 \times 3$ (RGB).
- **Block 1 (Conv1):**
 - Two convolutional layers with 64 filters.
 - Output shape: $32 \times 32 \times 64$.
 - Captures low-level features like edges and simple textures.
- **Block 2 (Conv2):**
 - Two convolutional layers with 128 filters.
 - After max pooling: $16 \times 16 \times 128$.
 - Detects corners and fine patterns.
- **Block 3 (Conv3):**
 - Three convolutional layers with 256 filters.
 - Output after pooling: $8 \times 8 \times 256$.
 - Captures object parts and medium-level features.
- **Block 4 (Conv4):**
 - Three convolutional layers with 512 filters.
 - Output after pooling: $4 \times 4 \times 512$.
 - Detects high-level features and object shapes.
- **Block 5 (Conv5):**
 - Three convolutional layers with 512 filters.
 - Output after pooling: $2 \times 2 \times 512$.
 - Encodes very abstract and semantic features.

- **Flatten + Dense Head:**

- Flattened output size: $2 \times 2 \times 512 = 2048$.
- Followed by custom dense layers for classification.